Liquidity Providing Validators: A BlueChip Upgrade

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Executive Summary

BlueChip is an on-chain economy, decentralizing subscriptions, and giving entities their own liquidity pool and token. This write up explores how BlueChip re-imagines delegated proof-of-stake (DPoS) so liquidity provision and rewards, not only raw token inflation and chain rewards, is the engine of validator and delegator rewards. By capping on-chain stake, redirecting excess capital into network liquidity, and adding an internal validator borrowing and lending protocol, BlueChip turns validators and delegators into active network liquidity providers.

Core Problems Addressed

- Unfair Yield for Liquidity Providers – Traditional AMMs dilute LP returns.
- Inflation-Heavy Validator Rewards – Reliance on token issuance erodes value and drives over-staking "arms races."
- Liquidity Volatility
- Delegator Penalties for Validator Misconduct – Conventional slashing cuts both validator and delegator stake, generally unfair to delegators.

Building Block	What It Does	Why It Matters
Stake Hard-Cap	Limits how much a validator can bond directly to the chain.	Prevents "whale" dominance and frees capital for liquidity.
Liquidity Vault	Smart contract where validators re-deploy surplus stake into chosen pools; yields LP NFT.	Provides extra liquidity to the network, gives validators differentiability without losing block-reward eligibility.
Composite Score	On-chain stake + Vault TVL; used as tie-breaker in validator set elections.	Rewards validators who deepen system liquidity.
Reserve Liquidity Fund (RLF)	Validator-only lending pool: 85 % base (bluechips) + 15 % bubble (pbluechips).	Gives validators short-term liquidity while keeping them economically accountable. Provides another revenue stream for validators.
pBluechip Wrapper	Interest-bearing layer on 15% "bubble" portion; wrapper may be burned on misconduct or off-chain transfers.	Enables validator diversification and dynamic trust scoring giving validators incentive to not act maliciously.
Reward-Withholding Penalty	Temporarily redirects a misbehaving validator's rewards to the RLF instead of slashing delegators.	Protects delegators and aligns incentives.

How the Design Solves the Problems

- Fair LP Compensation – Validators earn both block rewards and pool fees, boosting overall APY and making liquidity provision rational.
- Reduced Inflation Dependence – As Vault liquidity and trading volume rise, fee-based rewards substitute for minted tokens, allowing lower inflation targets.
- Price & Depth Stability – Continuous validator-provided liquidity dampens volatility across creator pools.
- Delegator Safety – Missteps trigger reward withholding and possible wrapper removal, leaving delegator stake untouched.
- Capital Efficiency – RLF loans cover withdrawal shocks; bubble assets are liquidated first, preserving core deposits and network stability.

Overview

This document briefly describes the premise of the BlueChip blockchain. More importantly, it looks into some of the core problems existing in blockchain and DeFi. The main problems looked into are: a liquidity provider's unfair yield, inflation as the main reward for validators, and the over-staking to counteract said inflation. BlueChip's proposal to solve these problems includes adding a wrinkle to classic designated proof of stake (DPOS). This wrinkle sets a firm limit to the amount a validator can stake directly to the chain. Any additional delegations or capital the validator accumulates can then be distributed to liquidity pools throughout the network by the validator's own Liquidity Vault. The validator is in control of their Liquidity Vault and which pools in the BlueChip network it supplies liquidity. By using capital to provide liquidity to pools in the network, the validator receives an LP NFT in return. The LP NFT is held in the Liquidity Vault. Validators, and in turn their delegators, receive rewards from the liquidity and the respective LP NFT.

By providing liquidity through the Liquidity Vault, validators increase their composite score. Composite scores consist of the validator's value staked directly to the chain, and value inside their Liquidity Vault. The primary number is the value staked directly to the chain, while the value in the Liquidity Vault is used as a tiebreaker mechanism. There are two checks carried out to finalize which validators provide stake to the chain. The first is a simple check. A simple check only looks at the amount staked directly to the chain. This check happens every 24 hours. The second check, a complex check, incorporates the validator's composite score and occurs every 5 days.

To support validator liquidity commitments and reinforce economic accountability, BlueChip also introduces the Reserve Liquidity Fund (RLF). The RLF is a decentralized lending protocol available exclusively to validators, allowing them to borrow up to 85% of their deposits. Validators borrow from the RLF based on short term liquidity mismatches which may occur if delegators withdraw their liquidity in times of liquidity constraint. In return for their deposits, validators earn yield based on the borrowing of other validators in the system. The RLF is composed of two parts: the base (85%) and the bubble (15%). The base is made of bluechips

while the bubble is pbluechips, a bluechip with a validator-denoted yield earning wrapper. Assets inside the bubble may be traded with outside assets, giving validators the opportunity to diversify their deposit as long as all trades to and from the bubble are at a 1:1 ratio. Repayments for RLF loans come from the validator's chain and liquidity rewards. If the validator is unable to cover the entire payment through rewards, their deposit begins getting liquidated starting with bubble assets followed by the deposit base.

This document explores the FINAL stage of the ecosystem. Each piece will be incrementally built and tested before being deployed. This will not be a one shot and done project. A timeline is listed at the end of this outline to provide further clarity on how this will come together.

Why Cosmos?

BlueChip's decision to build within the Cosmos ecosystem represents a strategic alignment with the most sophisticated blockchain architecture for application-specific chains. Here's why Cosmos provides the optimal foundation for a decentralized exchange and creator economy platform.

Core Infrastructure Advantages

1. Full Stack Sovereignty
   BlueChip maintains complete control over the entire technology stack, from consensus mechanisms to application logic. This sovereignty allows optimizing every layer specifically for a subscription-based DEX model, rather than adapting to the constraints of existing blockchain architectures.
2. Validation Control & Decentralization
   Unlike Layer 2 solutions that rely on centralized sequencers, BlueChip operates with a unique validator set, ensuring true decentralization where it matters most for a financial application. This approach eliminates single points of failure and censorship risks inherent in L2 architectures.
3. App Chain Thesis Alignment
   Cosmos pioneered the app chain model that represents the future of blockchain architecture. Rather than competing for blockspace with unrelated applications, BlueChip has dedicated computational resources optimized for trading and creator monetization.
4. Seamless Upgradability
   Cosmos also provides seamless on-chain upgrade paths through tools like Cosmovisor and governance-triggered upgrades. This capability ensures that BlueChip can evolve the protocol without disruptive hard forks, maintaining network stability while rolling out new features and improvements efficiently.

Technical Excellence

1. Instant Finality with Tendermint
   Tendermint's instant finality is crucial for DEX operations. Traders receive immediate settlement confirmation without waiting for probabilistic finality, eliminating MEV reorganization attacks and providing superior user experience compared to probabilistic consensus chains.
2. Native Interoperability via IBC
   The Inter-Blockchain Communication protocol enables true cross-chain liquidity without bridges or wrapped tokens. Assets from other Cosmos chains arrive as native IBC tokens, creating more trustworthy liquidity pools and reducing counterparty risk for users.
3. MEV Sovereignty
   Tools like Skip Protocol allow BlueChip to control Maximum Extractable Value distribution rather than losing it to external builders. This sovereignty enables MEV flows that benefit the community and creators directly.
4. Flexible Fee Economics
   Cosmos chains can implement custom fee models, allowing BlueChip users to pay transaction fees in any token rather than being restricted to the native asset. This flexibility improves accessibility and user experience significantly.
5. Optimized Performance Metrics
   Cosmos architecture enables high-performance throughput suitable for trading environments, with block times averaging 6–7 seconds and transaction capacity capable of scaling to thousands of transactions per second (TPS). These metrics provide the low-latency, high-throughput environment BlueChip needs to serve traders and creators efficiently.

Ecosystem Synergies

1. Sophisticated Validator Infrastructure
   The Cosmos validator ecosystem is already equipped to handle complex infrastructure requirements, making the validator-as-liquidity-provider model more feasible from day one. Validators understand the nuances of running financial infrastructure at scale.
2. No Blockspace Competition
   BlueChip's trading activity doesn't compete with NFT mints, gaming applications, or other unrelated use cases for throughput. Having dedicated blockspace ensures consistent performance during market volatility when traders need it most.
3. Governance-First Culture
   Cosmos chains expect active, sophisticated governance. This cultural alignment makes BlueChip's complex parameter adjustments and subscription model governance more natural and acceptable to the community.
4. Complementary Positioning
   Rather than directly competing with Ethereum DEXs, BlueChip can be strategically

complementary to existing Cosmos DeFi like Osmosis, focusing on a unique subscription economy niche while leveraging cross-chain liquidity.

Strategic Differentiation

1. Creator Economy Innovation
   Cosmos needs a creator-focused ecosystem, which BlueChip is uniquely positioned to deliver. The subscription-based model and creator monetization tools represent a new paradigm that makes more sense as a dedicated appchain than as smart contracts competing for attention on general-purpose chains.
2. Polaris Integration
   Polaris brings enhanced composability and interoperability features that will trump traditional L2 solutions. This positions BlueChip at the forefront of next-generation blockchain infrastructure.
3. Future Flexibility
   Should functionality beyond the core protocol be necessary, CosmWasm provides that capability without compromising specialized architecture.

Conclusion

BlueChip's choice of Cosmos represents more than technical preference, it's an architectural alignment with the future of decentralized applications. By building where sovereignty over the entire stack, instant finality for traders, and native cross-chain capabilities reigns supreme, BlueChip is positioned to deliver experiences impossible on other blockchain architectures.

The Cosmos ecosystem provides the perfect foundation for BlueChip to pioneer the intersection of decentralized finance and creator economy, establishing new standards for how blockchain applications can serve both traders and content creators in a unified, sovereign environment.

Definitions

The bluechip Token: The base token of the BlueChip protocol. Used for staking, delegation, lending via the RLF, and liquidity provisioning.

Liquidity Pool (LP): A pool of paired assets used for swaps and AMM trades. Validators may allocate funds to LPs in exchange for LP rewards, which are partially returned to delegators.

Proof of Stake (PoS): A consensus mechanism where validators are selected based on their staked tokens. In BlueChip, PoS is augmented by liquidity and performance metrics to dynamically influence validator rewards and responsibilities.

Delegator: A participant who delegates their tokens to a validator to earn staking rewards.

Unbonding Period: The window during which a delegator must wait to access withdrawn funds. This delay gives validators time to restore liquidity or tap the RLF.

Reserve Liquidity Fund (RLF): A protocol-native lending facility funded exclusively by validators. Validators deposit bluechips and receive pbluechips in return. They may borrow up to 85% of their collateral, and are subject to epoch-based repayments. The RLF also receives penalty fees and redistributed validator rewards in cases of protocol violations.

pbluechips: Wrapped versions of bluechip tokens deposited into the RLF. Each pbluechip contains metadata identifying the validator it belongs to and whether it is actively accruing interest. pbluechips can be traded, and are essential to tracking validator trust and collateralization.

Wrapper (Metadata): The smart-layer attached to pbluechips that encodes validator ID and active interest accrual status. Burned under malicious conditions, missed penalties, voluntary validator exit, or off-chain trades.

The Bubble: The 15% portion of a validator's RLF deposit that is not locked. It remains collateralized but is tradable, volatile, and can be liquidated. It impacts a validator's LTV and borrowing capacity, and is the first source used in penalty recovery.

RLF Base: The locked 85% of a validator's RLF deposit. This portion is not tradable or accessible until loans are repaid and the validator exits the RLF. It serves as the foundation of a validator's borrowing power and is liquidated only if bubble assets are insufficient.

Liquidity Vault: A smart contract or module that manages validator liquidity commitments. It allows validators to allocate funds to liquidity pools, handles unbonding requests from delegators, and enforces withdrawal grace periods to protect against sudden liquidity loss.

Liquidity Validator: A validator who allocates a portion of their stake to a Liquidity Vault and provides liquidity to liquidity pools in the BlueChip network earning liquidity rewards, but is currently not validating the chain.

Full Validator: A validator who is currently validating the chain. A full validator may or may not be providing liquidity to the Liquidity Vault and liquidity pools in the network.

Repayment Epoch: A fixed time interval of 3 days used to measure loan repayment periods, reward allocations, and penalty enforcement. Loan repayment schedules and validator penalties are calculated in terms of epochs.

Stake cap: The maximum possible score a validator can achieve by staking directly to the BlueChip blockchain.

Reserve Buffer: Collection of 2% of all reserve liquidity fund loan interest payments, penalty payments, and bubble-swap gas charges used as emergency reserve liquidity fund liquidity.

Composite Score: A metric used to determine validator standing based on their base stake, and liquidity pool exposure. It is used in complex validation checks to determine who is validating the network.

Simple Check: A validator eligibility check based solely on a validator's base stake amount.

Complex Check: A validator eligibility check that includes stake amount and liquidity pool exposure. Used for evaluating validators who participate in both validation and liquidity provisioning.

Base Loan Equation: The foundational formula for calculating the total amount a validator owes after taking out a loan from the RLF.

$$Base\ Loan = Principal + (Principal \times BPR \times LDR \times LF)$$

BPR = Base Payment Rate
LDR = Loan-to-Deposit Ratio Multiplier
LF = Liquidity Factor based on RLF utilization

Loan-to-Value Ratio (LTV): The ratio of a validator's outstanding loan to their total live collateral (base + bubble). Used to determine borrowing safety.

$$LTV = \frac{Outstanding\ Loan}{Base + Bubble\ Value}$$

LTV thresholds determine borrowing limits, freezes, and liquidation triggers.

Repayment Target (RT): The total amount a validator must repay over the course of a loan, including interest adjusted by loan duration and RLF utilization modifiers.

Validator Duration Modifier: A multiplier applied to the loan's interest based on the repayment length selected by the validator. Longer durations incur higher modifiers to compensate for repayment flexibility.

RLF Utilization Modifier: A dynamic multiplier that increases loan cost when overall RLF liquidity is low. Encourages faster repayment during liquidity stress.

Penalty Rate: The flat percentage used in calculating penalties for missed delegator withdrawals. Applies only to the missed amount, not the total delegation.

Offense Multiplier: A scaling factor applied to the penalty formula based on how many offenses a validator has committed. Designed to deter repeated violations.

Interest Redistribution: The process by which interest from a burned or stripped pbluechip is reallocated to the remaining active pbluechips in the protocol.

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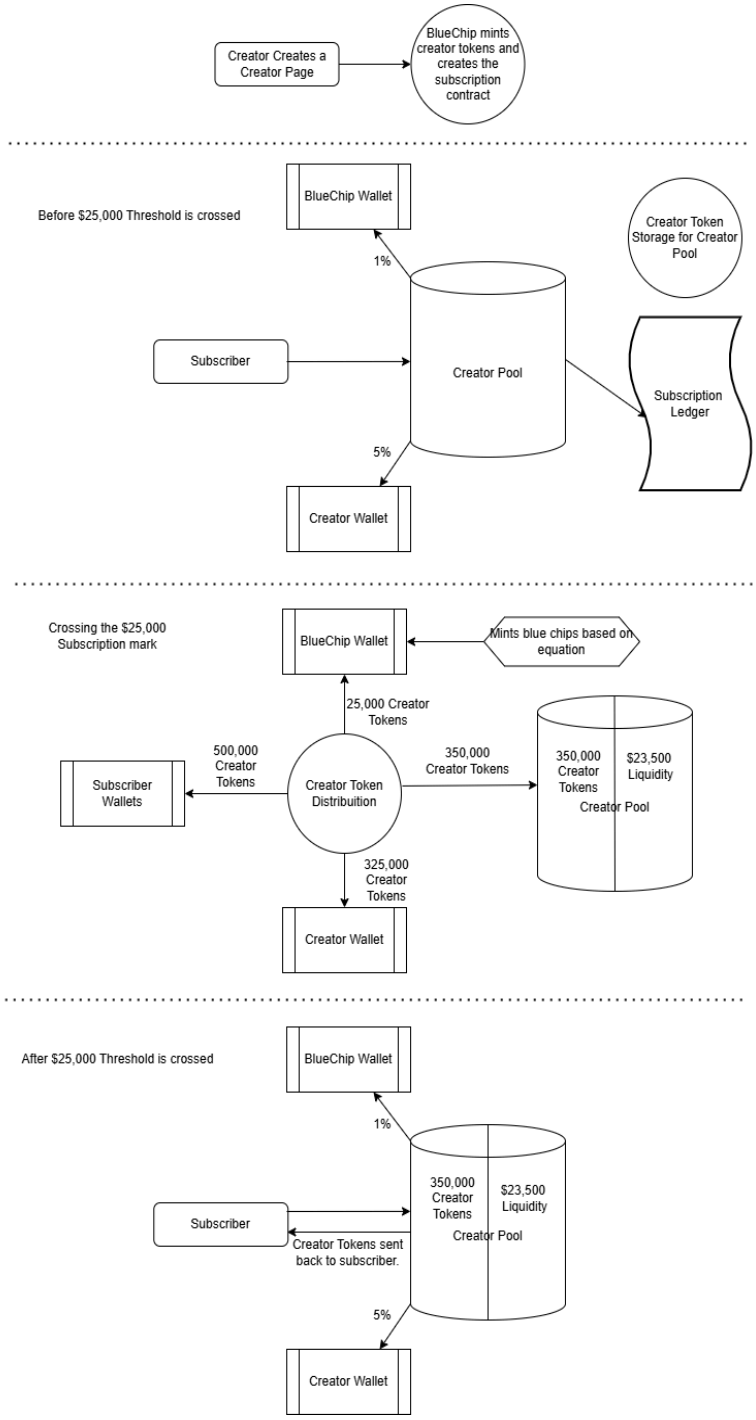
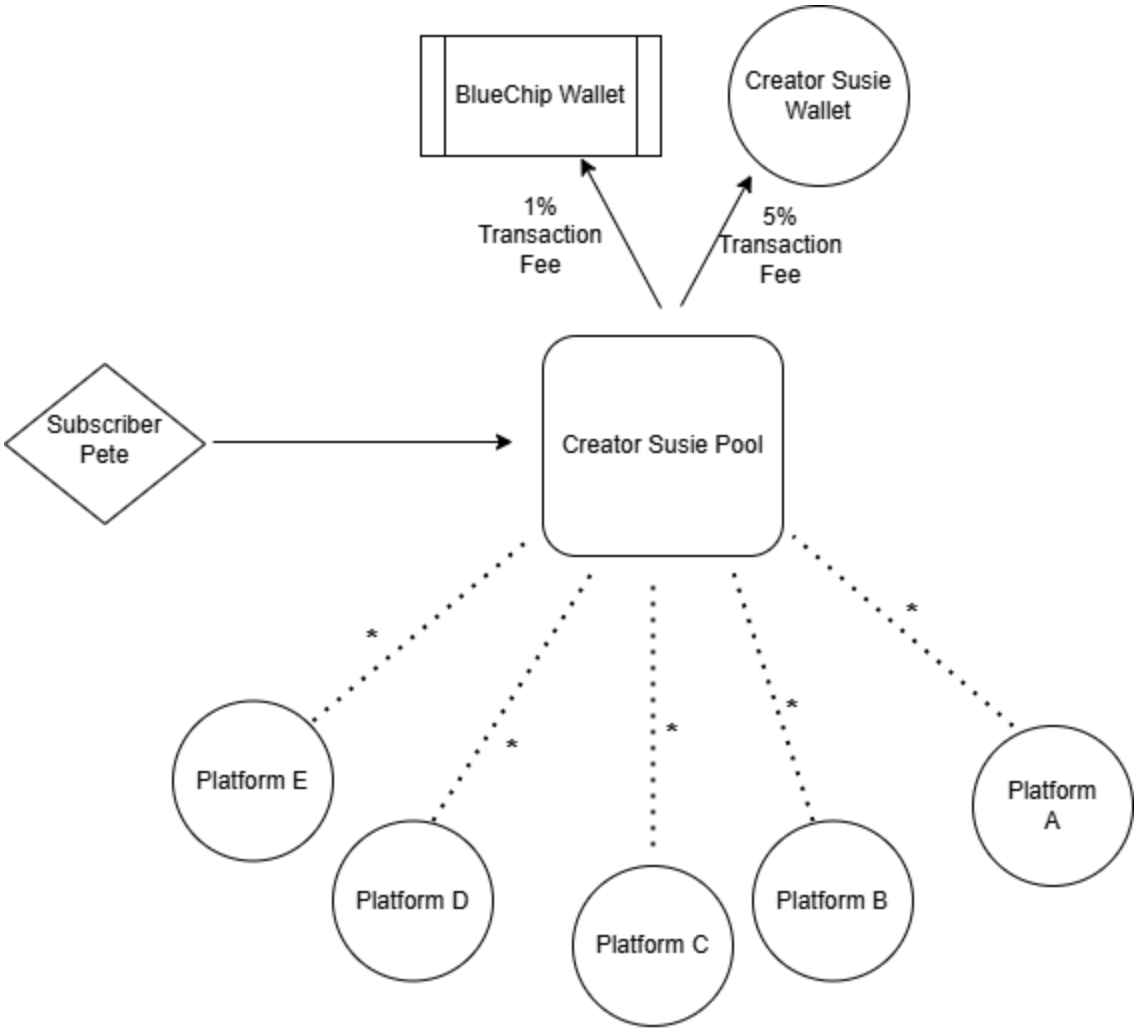
BlueChip Subscription Model

BlueChip is a blockchain network focused on creating a decentralized economy incorporating decentralized subscriptions for creators, businesses, or other entities. BlueChip does this by creating a liquidity pool for the entity with a special endpoint designating subscriptions. The following diagram displays how this happens. Note, BlueChip has developed a content creation platform so the following diagram is using creators as an example.



1. **Creator creates a page on BlueChip and verifies who they are.**
    a. BlueChip mints 1,200,000 creator tokens and hold them in a contract
    b. BlueChip creates a non funded liquidity pool allowing compatible wallets the ability to subscribe.
    c. Each transaction with the pool sends 5% to the creator wallet and 1% to BlueChip's wallet.
    d. A subscription threshold is created of $25,000 USD.
    e. Wallets can ONLY SUBSCRIBE to the creator pool. No other endpoints are available.
    f. A ledger is kept so the contract knows how much each wallet has contributed to subscriptions.
2. **Once creator receives $25,000 in subscriptions, crossing the threshold**
    a. 500,000 creator tokens are sent to the subscribers based on how much they have subscribed.
    b. 350,000 creator tokens are placed in the creators creator pool.
    c. 325,000 creator tokens are sent to the creator's wallet
    d. 25,000 creator tokens are sent to BlueChip's wallet.
    e. The money used for subscriptions minus fees, $23,500, is used to provide initial liquidity to the creator's liquidity pool .
    f. BlueChip mints blue chips based on the equation provided.
    g. Endpoints for simple swaps and liquidity provision are now available for the creators liquidity pool.
3. **Subscriptions after Creator reach $25,000**
    a. When a user subscribes to a creators channel, they transact with the subscription contract.
    b. This contract sends a 1% fee to BlueChip and a 5% fee to the creator.
    c. It then takes the remaining amount of the subscription cost and makes a swap with the creators liquidity pool sending the creator tokens to the subscriber.



* When subscribing to Creator Susie using the subscription contract formed on BlueChip, Subscriber Pete will gain access to Creator Susie Content on all other platforms that are synced with the Creator Susie Subscription contract, assuming Creator Susie has posted content there. Subscriber Pete will not be charged per platform, just the subscription price set by Susie in her subscription contract.

The above diagram also displays a main benefit for subscribers when subscribing using the liquidity pool endpoint. This benefit is the return of the entities (in this diagram it is labeled creators) designated crypto currency for their subscriptions. This effectively gives subscribers potentially free (immediately sell the tokens) or profitable subscriptions.

Due to subscription decentralization, the entity can connect multiple different platforms and websites to the same subscription. They are even able to create a subscription button on other platforms allowing them to accumulate subscriptions in multiple places:

One final thing regarding the subscriptions, the entity is able to make tiered subscriptions. In the example of creators, the creator can make free content, or content existing behind a paywall. Once a tier is made, the entity is able to add more, delete or edit the tiers.

Liquidity In The BlueChip Ecosystem

The BlueChip decentralized financial system relies heavily on liquidity pools and their ability to function properly by maintaining an adequate amount of liquidity. The current way pools receive liquidity is by actors, generally single actors, depositing value (known as liquidity) into them through the cryptocurrency pair established in the pool. This creates potential flaws including: **Impermanent Loss, Price Discovery, Liquidity Stability,** and more centric to the BlueChip chain, **the bluechip price**.

1. **Impermanent Loss:** Is when the prices of the assets in a liquidity pool changes relative to price of when they were deposited. It is technically "impermanent" because there is a chance the asset prices return to their original prices. In other words, if someone deposits liquidity and the liquidity pair is at a 1:2 ratio $10/$20 and 10 days later the pair changes to a 1:2 ratio $12/$24, an impermanent loss occurs. If the liquidity provider had held the assets they would have earned more (on paper) then having the tokens locked up as liquidity earning trading fees.

2. **Price Discovery:** The totality of a tokens price discovery includes multiple factors: liquidity pools, oracles, centralized exchanges and arbitrageurs. Liquidity pools obtain their prices by the assets internally weighing against each other inside the pool, oracles accumulate as much data from multiple sources in order to define a price, centralized exchanges select one or a few oracles to obtain their price information from, and arbitrageurs rover around making trades with liquidity pools, effectively bringing the oracle, centralized exchanges, and liquidity pool prices as close as possible while profiting on the trade differences (arbitrage trading).

   Liquidity pools can only update their price when a new block (new network state) is created and agreed upon (6 seconds on BlueChip). Unlike some chains, BlueChip does have instant finality (Cosmos SDK), nonetheless, oracle and centralized exchange prices are able to update in *real time* causing, by definition, an arbitrage. In other words an opportunity of impermanent loss. Effectively liquidity providers are paying arbitrageurs to recalibrate the pool's price to align with outside prices. They pay arbitrageurs with impermanent loss and collect fees based on the trade. Impermanent loss generally outweighs collected fees to the liquidity provider making it hard to rationalize providing liquidity to pools on a consistent basis.

3. **Liquidity Stability:** At a high level, liquidity stability is an asset's ability to be bought or sold within a liquidity pool without its market price being affected by the N+1 transaction of the asset. In simpler terms, the more liquidity, the more stable an asset's price inside a liquidity pool is.

4. **The bluechip price:** The bluechip derives its value through its use cases: staking with validators, providing liquidity to pools, and subscribing to different pools. The bluechip is *not a stablecoin*. Meaning it is subject to market dynamics. Why is this important?

   The bluechip makes up the 2nd half of all the entity liquidity pools. Since both token prices can independently change outside the pool, it causes interesting market dynamics and increased volatility. Its presence across all pools potentially stabilizes its price through interconnected market forces, reducing individual pool volatility. So as the network grows, there is a surface area like effect on its price. Each pool feeds off the other leading to a much more stable bluechip price.

5. **How these all work together:** The above four concerns work together in unison with liquidity stability as the bedrock. Having ample liquidity in the system greatly lessens the potential drawbacks of the other three. More liquidity equals better price stability for the bluechip, more resistance to impermanent losses, and less unfavorable price discovery for the pool. Moreover, as price stability, the bluechip price, and impermanent loss become more predictable, it incentivizes more people to provide liquidity and participate in the network through transactions.

Due to the importance of having a healthy amount of liquidity in the system, BlueChip views the liquidity pools in their system as an *extension* of the chain's security. To accommodate this, BlueChip has designed a system that enables validators to provide liquidity to pools in the network after meeting the chain's stake amount requirement. This proposal outlines how it works, potential issues of the system, and solutions to said issues.

<u>Problems Being Solved</u>

In relation to the service of providing liquidity, liquidity providers receive very little in return. Incentivizing token holders to provide liquidity is very important for the success of the BlueChip ecosystem. BlueChip aims to reward liquidity providers more accurately and remove negative trade offs stemming from providing liquidity.

Secondary Problem: Each token pair in the creator pools are made up of BlueChip's native token, the bluechip, and the entity's specific token. No stablecoins are being used. There needs to be ample and consistent liquidity to ensure everything stays relatively stable to ensure the best user trading experience.

Adjacent Problem: In delegated proof of stake systems, delegators are punished for validators behaving maliciously. For instance, if a validator goes down for too long, the chain carries out a slashing protocol burning a portion of the total staked value from the validator. Since the validator has delegators, the delegators are also punished, losing a share of their staked tokens as well. This feels unfair since most of these validators have close to 100% delegated value. In this proposed system, the idea of reward based punishment is introduced.

Inflation: As inflation becomes more of a hot topic, and blockchains at large become more efficient through clever solutions or raw technology, pure validation rewards are going to diminish. Without an alternative means for validators to earn meaningful rewards, network security could suffer.

Solutions To Problems

The solution to the primary and secondary problem revolves around validators. Validators can, but do not need to, redistribute their **over qualifying tokens** (any token amount above the required amount to be a validator) to their own **Liquidity Vault**, a smart contract that redistributes liquidity to liquidity pools in the ecosystem (validators themselves decide which pools receive liquidity). Bluechips submitted to the Liquidity Vault are counted towards the validator's "composite score". Composite score is the metric used to determine the amount of staking rewards a validator will obtain. The validator is not punished by losing staking rewards when redelegating bluechips to their Liquidity Vault vs staking them directly to the chain. This provides a secondary way liquidity enters the system while boosting rewards to validators and delegators. As the amount of liquidity in the system grows, the stability of the bluechip's price does alongside it.

Solutions to Adjacent Problem: This system introduces a reward withholding penalty for validators. This is not a universal implementation, however, any irresponsible or malicious behavior relevant to this outline, slashing is not involved. These actions are met by the validator losing future rewards for a period of time, while their delegators remain accumulating theirs. This way, the validators themselves suffer direct penalties for malicious or irresponsible acts, leaving their delegators unharmed.

Solution to Inflation: The BlueChip system addresses inflation by giving validators more nuanced, economically active roles. Validators are not just block producers, but also liquidity providers, and capital allocators. By providing these services, validators are rewarded above and beyond chain rewards, making validation viable to network validators. As the inflation rate lowers, participants feel more comfortable with leaving their assets outside of staking protocols and participating in the network. This also means, the more network participants, the more fees get directed back to validators.

Concerns to Address Regarding the Solutions

Implementing a system which gives validators this extra responsibility comes with potential risks. The first of these risks is finding sophisticated enough validators wanting to take on this burden. As validators become more professional in nature, this problem fades away. Professional validators want options to make themselves unique and strategically more optimal than their competitors. Moreover, in a delegator proof of stake system, true validator decentralization requires some form of meaningful differentiation. Think of a burger restaurant. If all of these restaurants had to sell McDonald's burgers, but McDonalds already offered the lowest price, what is the competitive advantage for these other restaurants? There really isn't any, and all business would centralize to McDonalds. Introducing new validator functionality, and even complexity, decentralizes validation.

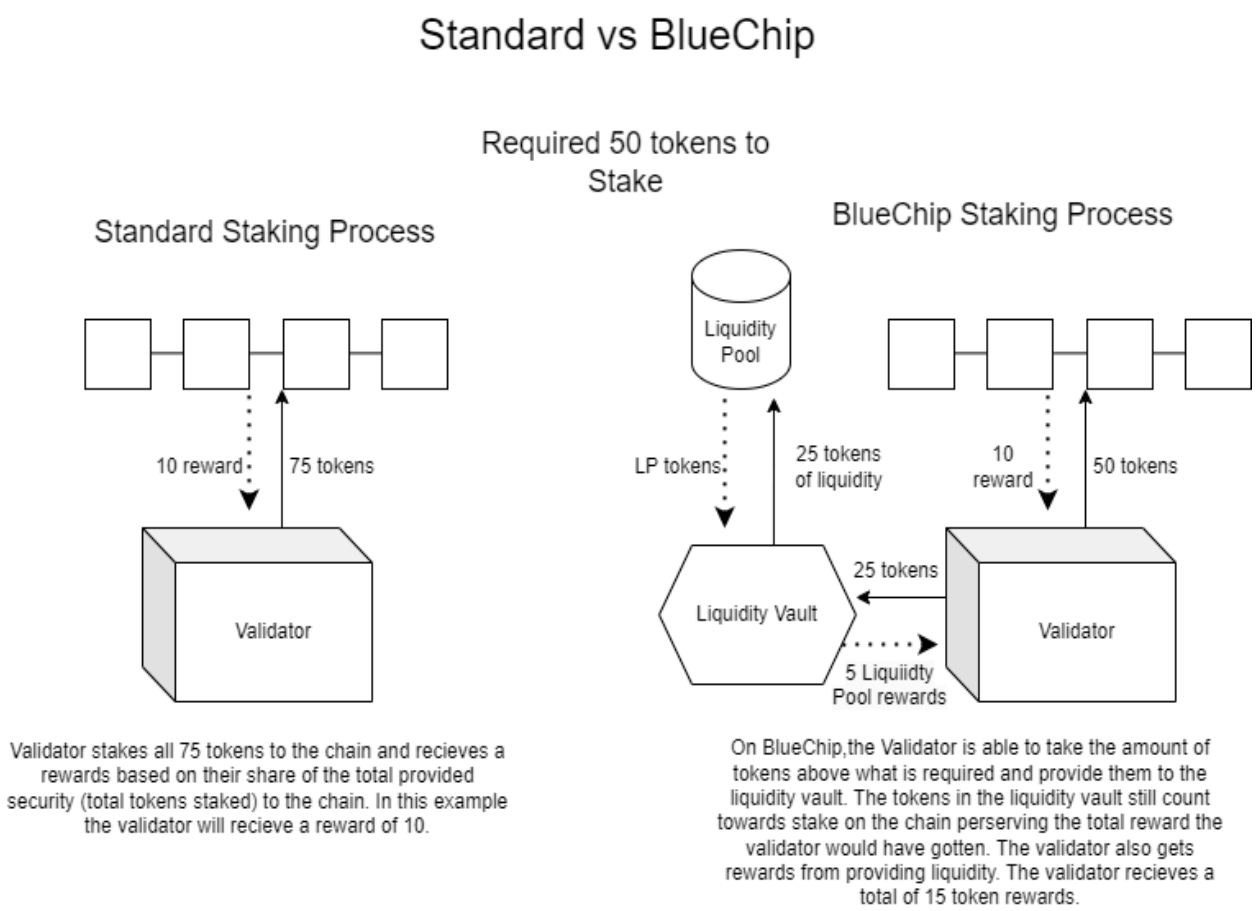
Second, concentrating an amount of liquidity to the validators can cause potential issues. Having this setup can either:

1. Centralize liquidity to certain pools.
2. Eliminate the single user liquidity provider.
3. Cause massive swings to certain pools' liquidity, primarily if 1 and 2 come to fruition.

It seems that issue 1 is easily solved due to market and competition between validators. 2 is a little more nuanced, but people still want to provide liquidity effectively making bets on their own. 3 is the real issue at play. 3 needs to be resolved through a few different chain standards, validator parameters, and market pressures.

A third risk is the fluctuating value of the LP NFTs held in a validator's Liquidity Vault. BlueChip is implementing a validator composite score and two types of checks: simple and complex. The simple check ignores the Liquidity Vault completely, and the complex check utilizes the median value of the validator's liquidity vault over a given time to help balance out the swings. Secondly, there are two types of validators in the system (**full validator**, **liquidity validator** explained below). Even if the validator is knocked off the chain's validating list (losing full validator status), they do not lose their Liquidity Vault or liquidity reward privileges, allowing themselves, and their delegators, to still accumulate rewards.

Finally is the risk of a validator liquidity squeeze. Validators are providing their resources, whether in the form of delegations or owned capital, to liquidity pools in the BlueChip system. It does not take one long to wonder, what happens when delegators pull their liquidity but the liquidity is tied up in the Liquidity Vault? There are a few tools being implemented aiding the validators in stressful moments like this.



<u>Liquidity Vault Specs</u>

After the Liquidity Vault provides liquidity to the specified liquidity pools in the network, the liquidity pool NFT is sent back to the Liquidity Vault to hold. The network considers LP NFTs in the Liquidity Vault when viewing the total amount staked by the validator. Rewards received from the LP NFTs are passed through to the validator and their delegators (based on parameters set by the validator), without the need of the validator to burn their position or any LP tokens.

Validator Parameters: Parameters designated with a **(V)** indicate mandatory voting when changed.

i. *Delegator unbonding time* - time period delegators must wait to receive their delegated bluechips back. The unbonding period is universally set and can only be changed if the validator has a deposit in the reserve liquidity fund. **(V)**
ii. *Validator commision*- what percent of validation revenue goes to validators vs delegators.
iii. *Validator liquidity pool commision*- what percent of the liquidity pool rewards goes to the validator vs delegators.
iv. *Liquidity Pools allocated to liquidity* - what pools are receiving liquidity from the validator.
v. *Liquidity in the Liquidity Vault* - How much liquidity the validator has locked up in network liquidity pools

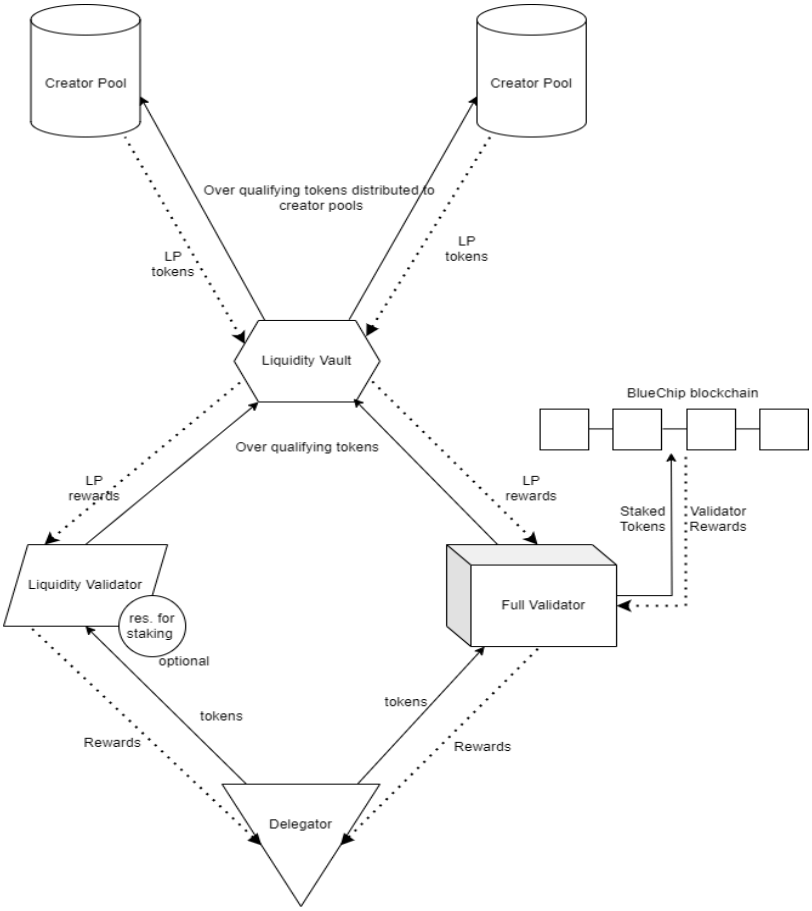
vi. *Liquidity Providing Change* - How long after the validator establishes it wants to relinquish its liquidity from a pool does it have to wait to receive the liquidity back (hard set by BlueChip).
vii. *Current additional funds* - the amount of side liquidity to ensure the validator can cover delegators withdrawals.
viii. *Over qualifying tokens* - the amount of delegated bluechips above and beyond the required amount to be a validator. Over qualifying bluechips are able to be sent to liquidity pools in the ecosystem or staked to the chain.
ix. *Dynamic Governance* - does the validator allow for delegator input on these parameters.
x. *Governance Lapse Time* - How long after implementing a change to their parameter set, does the change take effect for the validator node.

<u>Validator Types</u>

There are two types of validators in the network. **Full Validators** provide security to the chain and potentially to liquidity pools. They have a base amount of bluechips being used to secure the network and a potential amount of **over qualifying tokens** provided to liquidity pools through their Liquidity Vault. **Liquidity Validators** *only have over qualifying tokens*. In other words, they do not secure the chain, but they do utilize a Liquidity Vault providing liquidity to chosen pools earning liquidity rewards from their vault.

1. Full validators stake tokens, whether they are self owned or held through delegation, to the BlueChip blockchain. They receive token rewards as normal proof of stake validators would. In order to be a full validator, the validator must be a part of the validator set. Any additional bluechips held above the minimum validator set amount can be provided to the Liquidity Vault counting towards the full validator's composite score.

2. Liquidity validators are not currently validating the network through staking to the chain and are *not* part of the active validating set. Liquidity validators are simply participants accumulating assets to provide liquidity to the network's liquidity pools through their own Liquidity Vault. Depending on their goals, they may choose to preserve the minimum stake amount of bluechips rather than placing all assets in the Liquidity Vault enabling them to start validating the chain when their composite score becomes high enough.



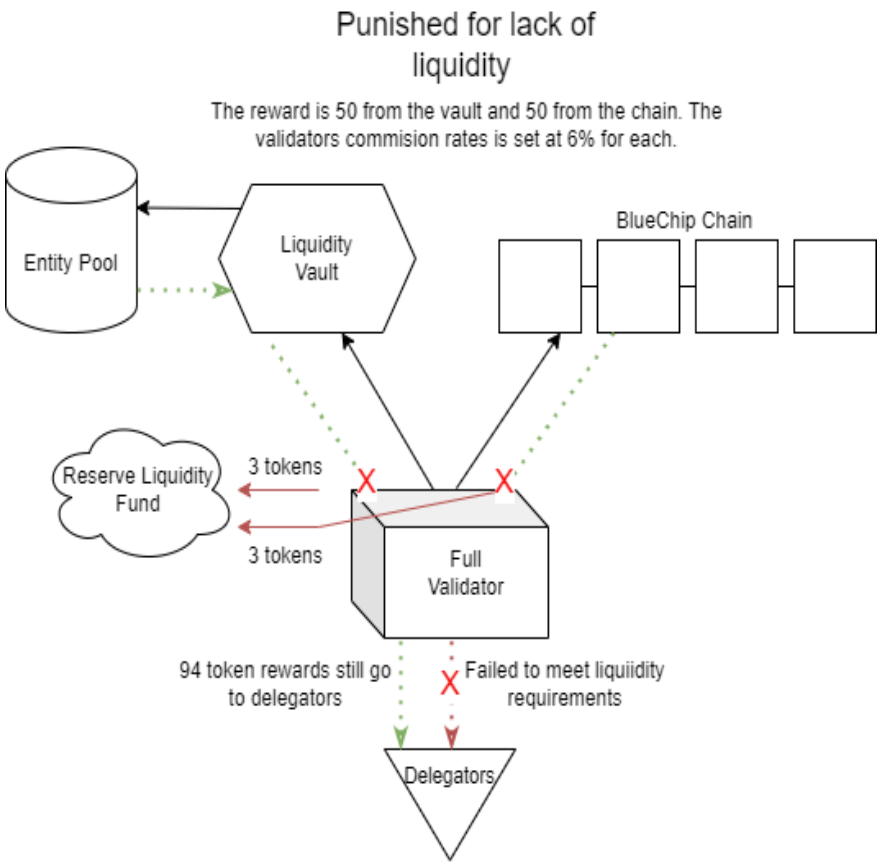
Liquidity Vault and Penalties

The Liquidity Vault is an extension of the validator and can only be accessed by the validator it is designated too. The validator sends a transaction to the vault with an amount of bluechips and designates which liquidity pool(s) it would like to supply liquidity to. BlueChip is adopting a NFT framework to handle liquidity pool tokens. Instead of receiving individual LP tokens, which generally require burning in order to retrieve rewards, users can now simply collect fees from their NFT incrementally. When wanting to retrieve their provided liquidity, the validator sends a transaction to the Liquidity Vault stating which liquidity and how much it would like to withdraw. If the transaction removes all the liquidity, the NFT is burned. After initializing this call, there is a grace period before the liquidity is removed from the pool. The grace period is to protect pools against massive liquidity swings. **Liquidity providing change** is set by BlueChip and is universal across all validators. Once the tokens are withdrawn from the liquidity pools, they *no longer count towards the validator's composite score*. The validator must stake the tokens to the chain, or provide liquidity to a new pool manually.

The Liquidity Vault can override the parameters if there is any disagreement. For instance, let's say a validator sets a withdrawal period of 10 days and they are heavily allocated to liquidity pools. Unfortunately, a significant part of their delegator base decides to withdraw their money. There is now a 10 day grace period for the validator to come up with the funds to supplement

their allocation otherwise the withdrawal by the delegator is denied by the vault.

To punish the validator for not having liquidity for a delegator exit, the validator loses a portion of future validator commissions and LP rewards unless they can pay off the penalty directly. The validator losing rewards has no impact on its delegators ability to continue to accumulate their rewards. The rewards lost by the validator are placed in a protocol named, the **Reserve Liquidity Fund**.



Punishments for not having liquidity ready for your delegators is as follows:

Offense #	Offense Multiplier
1	1.1
2	1.2
3+	1.5

$$Total\ Penalty = Missed\ Withdrawal\ * (.05 * Offense\ Multiplier) + 500$$

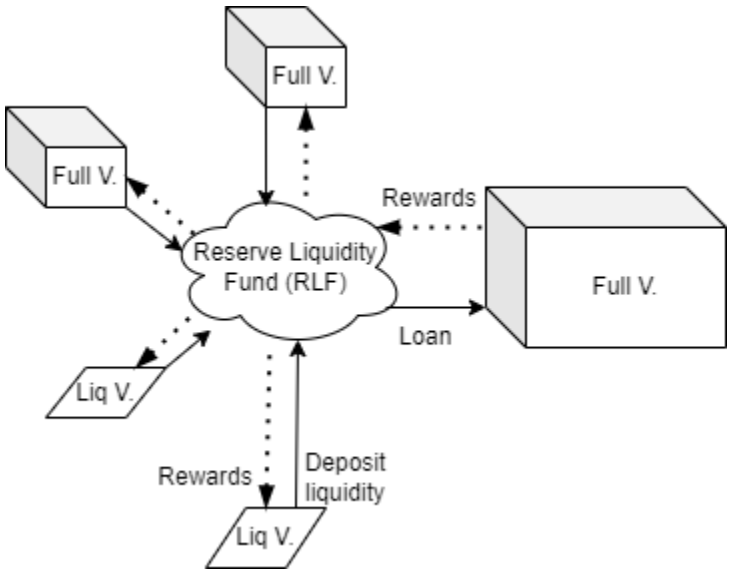
<u>The Reserve Liquidity Fund</u>

The **Reserve Liquidity Fund (RLF)** is a protocol validators can contribute bluechips to. Validators can borrow from the RLF when they are in need of liquidity (delegators begin pulling funds and they can't cover all of it). The RLF pays fees out to all other validators based on their allocation. The RLF is highly liquid for lending and only validators have access to it. Validators can only borrow from the RLF if they first make a deposit into it.

a. The validator makes payments on their loans using their rewards. These payments again do not affect delegation rewards for the delegators. Validators are also able to use **current additional funds** to pay off these loans.
b. Generally speaking, the RLF would only be used where there is no current additional funds available for the validator. So essentially what the RLF use case boils down to is this:

LP rewards + RLF rewards + certainty to recoup
$>$
RLF fees + uncertainty of delegators staying power

If a validator can consistently achieve this, it makes sense for the validator to put ample liquidity into the RLF or the Liquidity Vault. In the event a validator has set a low delegator unbonding time, making delegations more liquid than their staked currency, the validator would simply add *chain rewards* to the left side of the arrow and a much heavier weight to *uncertainty of delegator staying power.*



c. A validator can only utilize the RLF for loans if they first deposit funds into it. The amount a validator can take as a loan is directly correlated to this amount. The amount a

validator can take out as a loan is 85% of their total current deposit in the RLF. The deposit into the RLF must come from the *validator themselves.* The validator can not use delegated funds for the RLF. The RLF is purely for validators and is fully funded by them as well. If a validator could use delegator funds and provide them to the RLF, it would in some sense nullify the purpose of the collateral to begin with. Secondly, if delegators are afraid validators may do something shady with RLF loans, they may either be hesitant to take their liquidity out, or would be afraid to deposit with validators to begin with.

d. In a severe circumstance where the validator is a debtor to the RLF and they decide to stop validating the network through liquidity and chain security, the RLF recoups lost funds from the validator through the validator's RLF deposit. If the validator loses full validator status only, but continues to earn rewards through their Liquidity Vault, those rewards are used instead. In any circumstance where rewards are not enough to cover the full loan payment (or penalty), the remainder of the payment is covered by the RLF deposit. If the RLF deposit is dwindled to zero, the validator's unbonding period for delegators is reset to the universal standard.

<u>RLF Base Loan Calculation</u>

Below is how the RLF loans work:

i. First some definitions and outline reasonings.
   1. *Principal (P)* - The initial amount borrowed.
   2. *Base Payment Rate (BPR)* - The initial multiplier any loan from the RLF takes on. This is simply to set a floor for loans against the RLF.
   3. *Liquidity to Deposit Ratio (LDR)* - The larger the loan relative to the validator's deposit, the higher the additional interest on the loan.

Loan Type	% of Deposit	Multiplier
Safe	50%	1.00
Moderate	50% - 75%	1.10
Risky	75% - 85	1.20

   4. *Liquidity Factor (LF)* - Payment rate adjusted for available liquidity. The lower the available liquidity in the RLF, the higher the payment rate is on a validator's loan.
   5. $LF = 1 + (\frac{Total\ Liquidity - Available\ Liquidity}{Total\ Liquidity})$

ii. The final equation for a loan from the RLF is as follows:

*Base Loan = Principal + (BPR * LDR * LF)*

iii. Example for Base Loan:

1. Total Liquidity in the RLF: 10,000,000 bluechips
2. Available Liquidity: 7,500,000 bluechips
3. Validator RLF deposit: 200,000
4. Loan Requested: 100,000 bluechips
5. Loan to Deposit Ratio: 1.00
6. Base Payment Rate: 2%
7. Liquidity Factor (LF):

$$LF = 1 + \left(\frac{10{,}000{,}000 - 7{,}500{,}000}{10{,}000{,}000}\right) = 1.25$$

$$Base\ Loan \approx 100{,}000 + (100{,}000 * 2\% * 1.00 * 1.25) = 102{,}500.00$$

<u>RLF Repayment Structure</u>

The RLF loan repayment structure depends on two factors: the duration of the loan set by the validator and the current usage of the RLF in *total,* not just specific to the validator taking out the loan. The equation to determine the Repayment Target (RT) is as follows:

$$x = Loan\ Interest\ * \left(\frac{(Validator\ Duration\ Modifier + RLF\ Utilization\ Modifier)}{2}\right)$$

Repayment is denoted in Epochs which are 3 days long. To find the minimum payment for the validator's loan, simply add the above to the base loan and divide it by the total number of epochs of the loan.

$$RT = \frac{(Base\ Loan + x)}{Loan\ duration\ in\ Epochs}$$

If a validator is unable to provide the minimum amount for their payments, the RLF liquidates a correlated amount from the validator's RLF deposit, beginning with **the bubble** (explained below), to compensate for the missing payment. Below are the two tables that decide the Validator Duration Modifier and the RLF Utilization Modifier. Once decided, the repayment amount for the loan is locked in and does not get adjusted for market forces.

Validator Duration Modifier

Loan Type	Duration	Duration Modifier	Use Case/Notes
Liquidity Loan	10 Days (3 Epochs)	.85	Short-term liquidity fix (e.g. unbonding gap coverage)
Quick Loan	30 Days (10 Epochs)	1.00	Standard fast access to funds
Medium Loan	60 Days (20 Epochs)	1.10	Flexible repayments for moderate needs
Long Term Loan	120+ Days (40+ Epochs)	1.35	Extended repayment for strategic borrowing.

RLF Utilization Modifier

RLF Utilization	Utilization Modifier	Repayment Pressure	Notes
0-39%	1.00	Low	Plenty of available liquidity
39% - 59%	1.10	Moderate	Normal Usage
59% - 79%	1.20	High	Liquidity Tightening, quicker repayment encouraged
79% - 90%	1.35	Very High	RLF under stress
90% - 100%	1.55	Critical	Emergency Tier, aggressive repayments encouraged.

<u>Full RLF Loan Example</u>

Here is an example of what a loan repayment schedule may look like:

Validator RLF Deposit: **200,000 bluechips**

Loan Requested (Principal): **100,000 bluechips**

$Total\ Loan \approx 100,000 + (100,000 * 2\% * 1.00 * 1.25) = 102,500$

Loan Type: Medium Loan (**60 days / 20 epochs**)

Validator Duration Modifier: **1.10**

RLF Utilization: **72%**:

RLF Utilization Modifier: **1.20**

$$RT = 2,500 * (\frac{(1.1 + 1.2)}{2}) = 102,875$$

$$\frac{102,875}{20} = 5,143.75\ bluechips/epoch$$

In summary, the validator has to make a payment of 5,143.75 bluechips every 3 days (epoch) to repay their loan. It is important to remember, the validator effectively gains control of the liquidity the delegator is withdrawing after the undelegation period. Therefore, this repayment becomes more flexible and exponentially easier to make on time.

e. Applying the above penalty vs. loan trade off to a potential real world validator circumstance:
    i. Annual BlueChip chain rewards: 5,256,000
    ii. # of validators: 50
    iii. Total liquidity supplied by validator: 500,000
    iv. Validator commission (chain rewards): 5%
    v. Validator liquidity pool commision: 50%
    vi. Total chain rewards: 105,120 * 5% = 5,256
    vii. Liquidity pool rewards (assumed APR 20%): 50,000

viii. Total rewards for validator: 55,256 BlueChips
ix. % of rewards as loan: 5.2%
x. Would be penalty:

$$Total\ Penalty = 100,000\ *\ (.05\ *\ 1.1) + 500 = 6,000$$

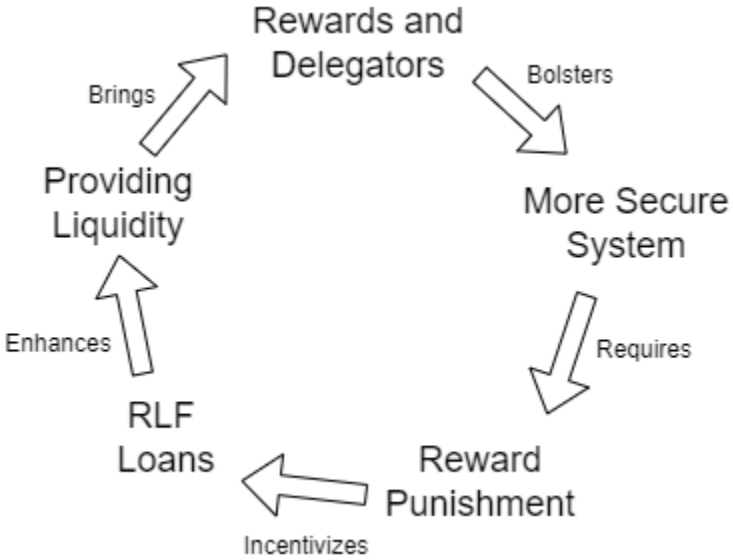
xi. There are a few assumptions made for this example.
    1. One, it is safe to say the validator wants a larger percentage of their liquidity pool rewards as they are taking on more risk. This makes even more sense as these rewards are *above and beyond* standard chain rewards for delegators making the lopsided margins less bothersome for them. Ultimately, market forces dictate the best number here. 50% seems like a relatively safe assumption.
    2. Second, a 100,000 loan is 20% of this validator's total stake amount. This is a large sum for this particular validator. Even if a single party owned 20% of the validator's total delegations, they would most likely not pull out 100% of it at once granted its relative size.
        a. Thirdly, the loan equates to 5.2% of the validator's annual rewards or 26% of the annual rewards they received from this proportion of their stake. Comparing this to slashing, a 5.2% penalty is low while 26% is very high. However, this is theoretically based on rewards. Meaning, it is more of a tax on what the validator is earning vs slashing which is effectively reaching into the validator's bank account and taking a percentage of what they own (or have delegated to them).
        b. Finally, these penalties do not affect the delegators stake. Delegators not being harmed by penalties incentivizes them to seek riskier validators. In turn, they are more likely to understand and accept liquidity-related risk.

f. RLF loans are designed to not have compounding interest. They are simply a flat rate the validator is required to pay back. This strategy was decided as having validators enter a potential debt death spiral is dangerous for the ecosystem. Secondly, as RLF loans effectively use validator rewards as collateral, it would add a layer of unnecessary complexity to add compounding interest.

g. Although the RLF is highly liquid for lending, there is a universal unbonding period for bluechips providing liquidity to the RLF. This gives borrowers more certainty on available liquidity.

The relationship between RLF loans, validator penalties, and liquidity provision is critical to the system. It is designed to incentivize validators to supply liquidity to the pools without imposing

excessive penalties. If RLF loans were too severe, validators would become overly risk-averse, which could undermine the system's overall benefits. More provided liquidity brings higher rewards for validators, which makes it more secure.



<u>The RLF Bubble and pbluechips</u>

The RLF behaves like a decentralized lending protocol. Depositors deposit, and borrowers borrow. As stated above, a validator can only borrow 85% of their RLF deposit. The additional 15% (the percentage above the available 85% for loans) is able to be traded, leaving the base 85% locked into the protocol. The 15%, known as **the bubble**, still maintains the function of being collateral and can be liquidated when the loan outstanding becomes debased or the validator holding the loan behaves maliciously.

When a validator deposits into the RLF, they do not receive new tokens in return for their full deposit (like an aToken, cTokens etc). Instead their deposited bluechips get wrapped in an interest bearing wrapper becoming pbluechips. This wrapper is handled in the metadata of each bluechip. The metadata is simple and provides insight on which validator the pbluechip is tied to, and whether or not the pbluechip is actively accumulating interest.

Each validator has its own identifier for their wrapper so buyers, holders, and the protocol know which validator's liquidity is being held. A pbluechips wrapper is burned when the corresponding validator behaves maliciously, does not payback appropriate penalties or loans, or finally, if the validator withdraws their liquidity from the RLF. By designating pbluechips to validators, and burning the wrapper when the corresponding validator behaves maliciously, it creates a market for their pbluechips based on trust, incentivizing good behavior. Wrappers are burned at random and there is no FIFO or LIFO reasoning.

When trading bubble assets, all trades to and from the bubble must have equal value on both sides. A validator can not sell assets for a total of $1 and receive a deposit into the bubble for a total of $1.50. All trades must be 1:1. It is important to note, upon deposit, the pbluechips in the bubble are valued simply as bluechips. The protocol only considers the premium tagged to pbluechips when they enter the bubble from outside sources or when the validator is selling the pbluechips to accumulate other assets.

If a validator does trade their pbluechips out of the bubble, and the assets collected depreciates in value, the validator can now borrow less from the RLF than they could before. This is true as well for the opposite scenario. As stated above, all deposits are treated equally. Premium pricing is not applied until the pbluechips are being traded out of the bubble. If a validator is highly trusted and their interest wrapper carries weight amongst the community, they are able to sell their pbluechips at a premium, growing their borrowing capabilities immediately. Validators are able to hold other validator pbluechips in their bubble.

If the bubble experiences rapid devaluation (e.g., due to a market crash), the validator's live collateral value drops accordingly. Since the RLF calculates Loan-to-Value (LTV) ratios in real time, a falling bubble value pushes the validator's LTV upward. Once the LTV exceeds 90%, the validator's position is automatically flagged for liquidation. The RLF begins reclaiming funds by liquidating bubble assets first, preserving the base deposit if possible. This protects the protocol's solvency without needing to interpret market movements — LTV enforcement is automatic and agnostic to cause.

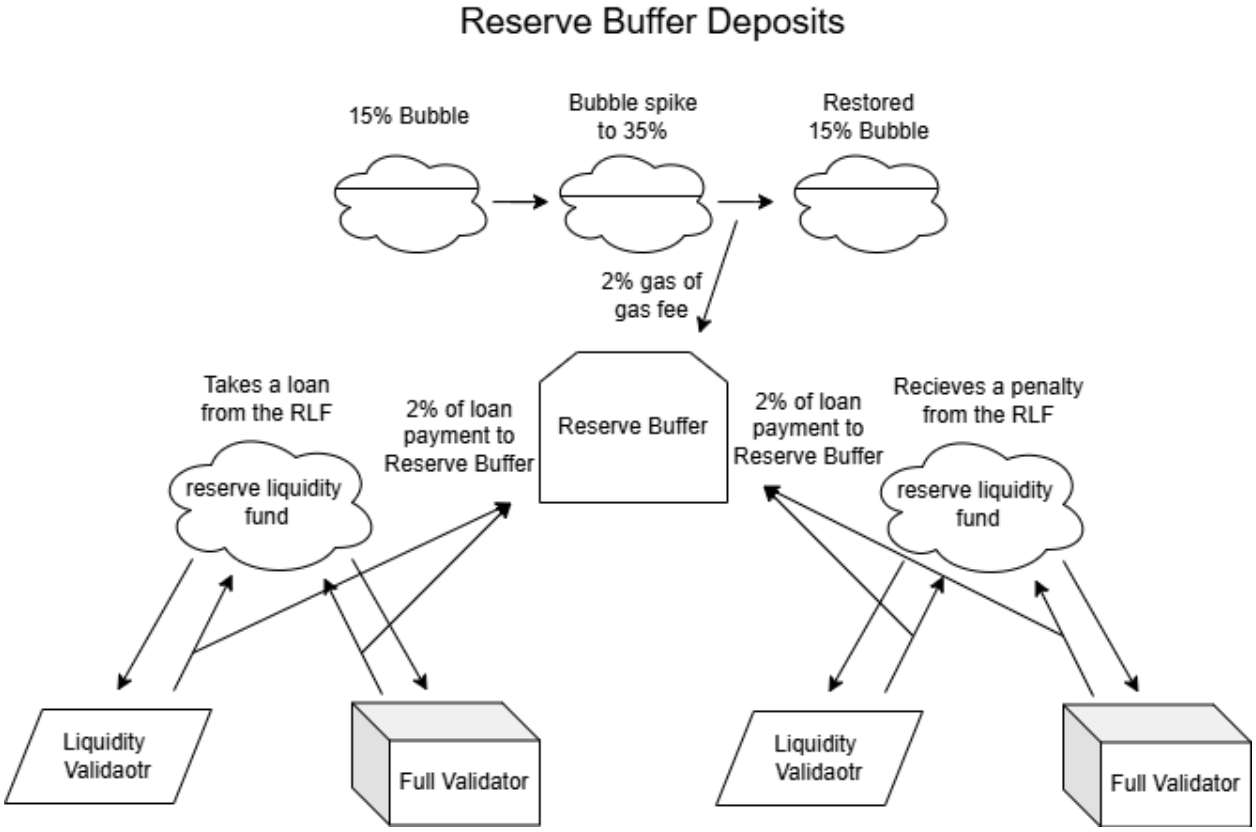
Loan-to-Value

Condition	**Action**
LTV ≤ 85%	✅ **Safe**
LTV 85%–90%	⚠️ **Warning, borrow freeze**
LTV > 90%	❌ **Liquidation Begins**

Original Deposit	Locked 85%	Bubble Value	Total Live Collateral	Outstanding Loan vs Max Loan	Loan to Value (%)	Status
1000	850	150	1000	850:850	85%	✅Safe
	850	250	1100	850:935	77.27%	✅Safe
	850	140	980	850:833	86.7%	⚠️Frozen
	850	30	880	850:748	96.6%	❌Liquidation

A spike in appreciated value in bubble assets, followed by a validator loan, followed by a severe correction and devaluation of the bubble assets, causes major issues for the RLF. The RLF is left heavily over collateralized and with specific market events, the entire system would crumble due to lack of repayment. To help mitigate this, there are incremental rebalances between the bubble and the base collateral. Every epoch any value accrued in the bubble above 17% gets swapped into bluechips and deposited in the base collateral securing the validator's gains, recalibrating the bubble to 15%. The swap is executed by the module and the gas fee is debited from the validator's gains. Secondly, If the bubble ever appreciates to over 30% of the total deposit (say a massive spike in a specific bubble token), the RLF automatically swaps the gains down to the bubble equaling 20% and the base collateral equaling 80%. Then of course the bubble and base collateral balances would be rebalanced to 15:85 respectively, on the next epoch.

To further ensure a healthy RLF, a small 2% fee is skimmed from interest payments, penalty payments, and bubble-swap gas charges. This fee is placed in a non-lendable reserve account called the reserve buffer. This collection of fees behaves as a backstop for the RLF in case of a mass downturn or liquidation event. Funds in the reserve buffer may only be used to cover shortfalls after bubble-first and base-second liquidations have been executed. If the reserve buffer reaches 10% of all RLF deposits, the fees are redistributed to staking rewards.

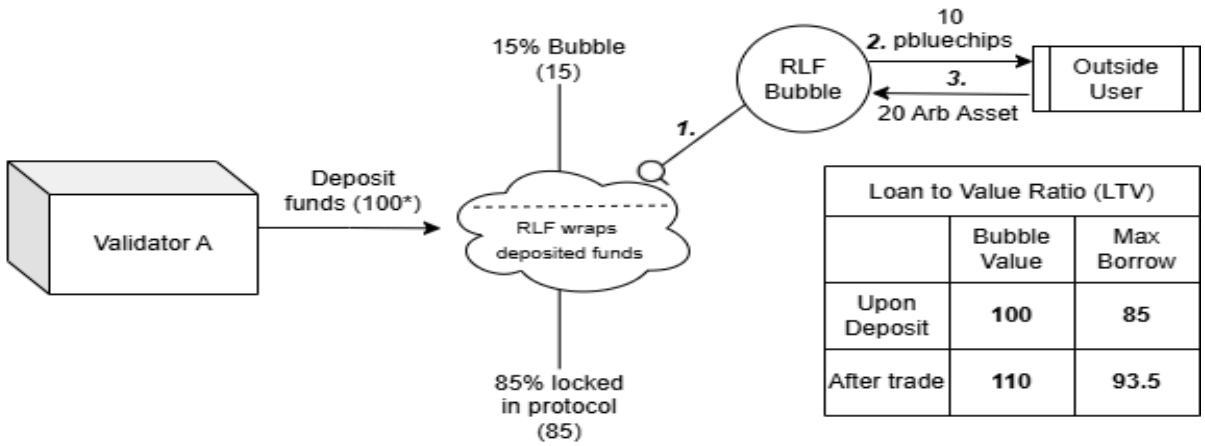


When withdrawing from the RLF, the validator liquidates an equal percentage from their bubble and their basis retaining the current ratio. For instance, if a validator has a total value of 100 in

their RLF and the current ratio is 85:15 in favor of the basis, and they withdraw a value of 15 from their RLF deposit, they withdraw:

$$15/100 = (.15 * 85) + (.15 * 15) = (12.75basis) + (2.25bubble) = 15\ total.$$

Since pbluechips are denoted with unique metadata, they are not interoperable with other chains in the Cosmos ecosystem. If a pbluechip gets traded off of the BlueChip chain, the metadata is stripped causing the pbluechip to revert back to being a standard bluechip and all its future interest payments are dispersed to the other pbluechips in the ecosystem. This does not prohibit the individual bluechip from becoming a pbluechip if a validator were to redeposit it into the RLF at a later time. Bluechips themselves are of course able to be traded across the cosmos ecosystem.

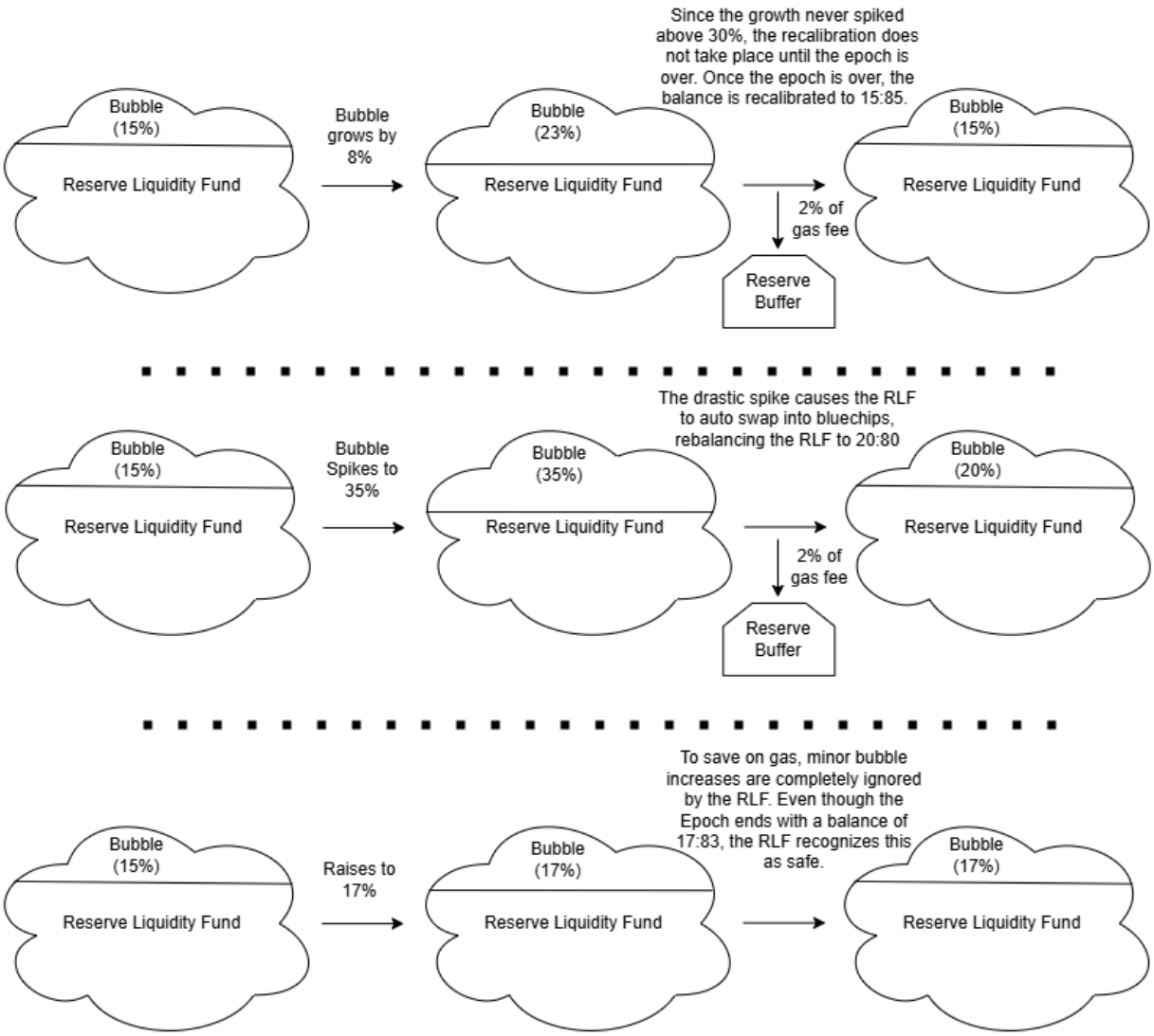


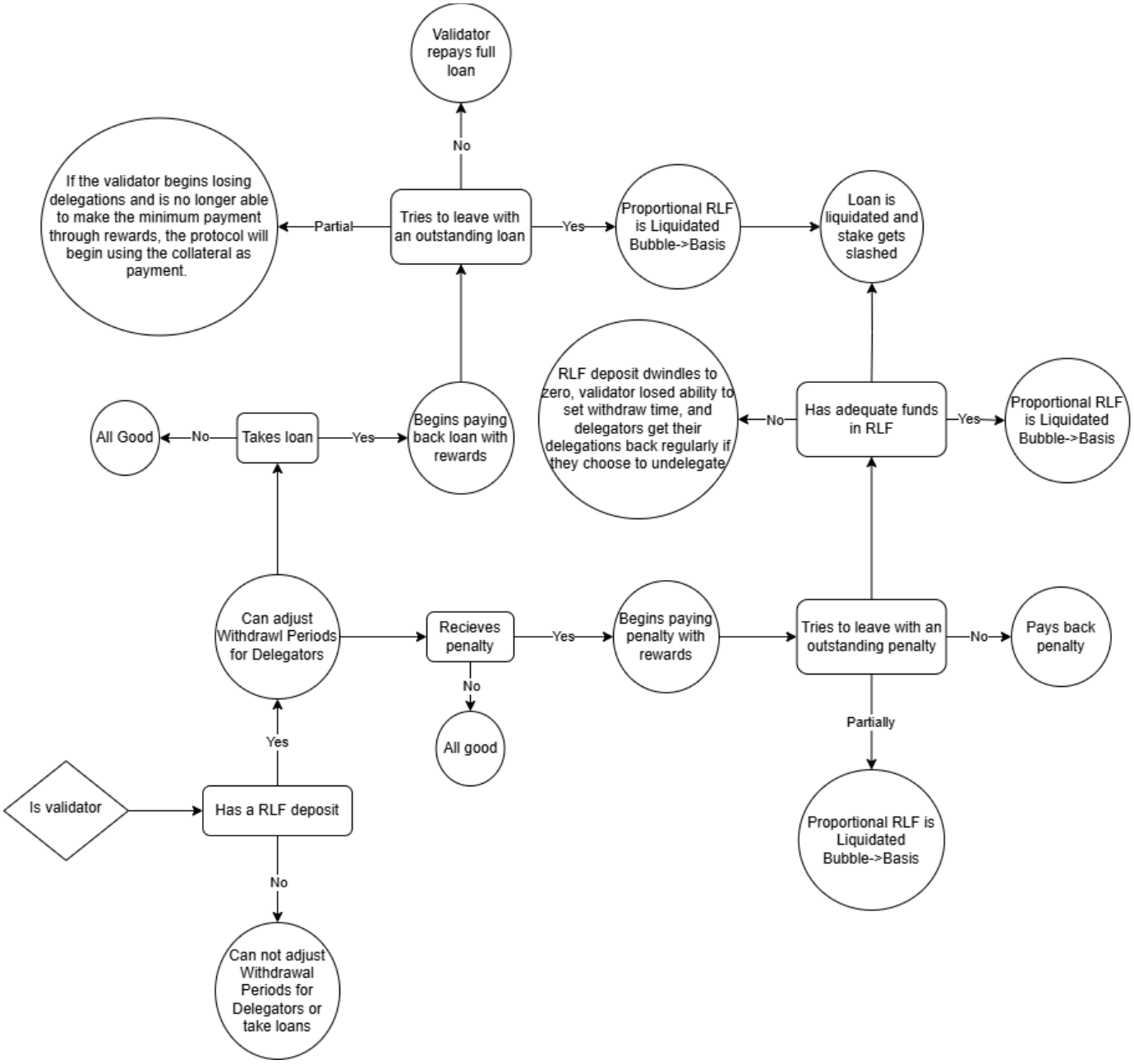
Loan to Value Ratio (LTV)		
	Bubble Value	Max Borrow
Upon Deposit	**100**	**85**
After trade	**110**	**93.5**

1. On deposit, pbluechips are valued as standard blue chips while idle in the RLF bubble

2. Although pbluechips are valued as blue chips while idle, they will be valued correctly when trying to trade out of the bubble.

3. All trades with the RLF must be of 1:1 value. The RLF can not recieve or send tokens unless traded.


Since the growth never spiked above 30%, the recalibration does not take place until the epoch is over. Once the epoch is over, the balance is recalibrated to 15:85.
Bubble (15%)
Reserve Liquidity Fund
Bubble grows by 8%
Bubble (23%)
Reserve Liquidity Fund
2% of gas fee
Reserve Buffer
Bubble (15%)
Reserve Liquidity Fund
The drastic spike causes the RLF to auto swap into bluechips, rebalancing the RLF to 20:80
Bubble (15%)
Reserve Liquidity Fund
Bubble Spikes to 35%
Bubble (35%)
Reserve Liquidity Fund
2% of gas fee
Reserve Buffer
Bubble (20%)
Reserve Liquidity Fund
To save on gas, minor bubble increases are completely ignored by the RLF. Even though the Epoch ends with a balance of 17:83, the RLF recognizes this as safe.
Bubble (15%)
Reserve Liquidity Fund
Raises to 17%
Bubble (17%)
Reserve Liquidity Fund
Bubble (17%)
Reserve Liquidity Fund



Validator Action	Protocol Outcome
Deposit into RLF	85% as base 15% into tradeable bubble
Pays loan on time	No penalty; maintains trust & interest accrual
Misses loan repayment	Bubble assets liquidated first, then base if needed
Misses delegator withdrawal	Penalty applied; rewards withheld

Trades out pbluechips at loss	LTV increases; borrow limit drops
Bubble value crashes	LTV may breach; triggers borrow freeze (85%) or liquidation (>90%)
Trades out pbluechips at premium	LTV decreases; borrow limit increases
Bubble value increases	LTV decreases, higher borrow limit - current payments remain
Withdraws RLF deposit	pbluechip wrapper is burned
Move pbluechip off-chain	Wrapper stripped; interest redistributed
Re-deposits stripped pbluechip	Assigned new wrapper with new validator ID

RLF Circuit-Breaker Safeguard

Per-validator "freeze" thresholds stop any single borrower from over-leveraging, but a handful of large validators ("whales") can still freeze simultaneously during a correlated crash. Their outstanding loans remain on the books and can push the Reserve Liquidity Fund (RLF) utilisation dangerously close to 100 %, starving liquidations and redemptions. A circuit-breaker (CB) is a simple, protocol-level kill-switch that:

1. Halts new borrowing the moment aggregate utilisation or system-wide LTV crosses a danger line.
2. Buys one epoch of breathing room for liquidations, repayments and governance to work, preventing a reflexive death-spiral.
3. Stays completely inert in normal conditions (zero gas overhead, negligible audit surface).

pbluechip Pricing

A validator's pbluechip price is derived from its Trust Score, which reflects its recent on-chain behavior and capital management. The Trust Score is updated periodically and feeds directly into a price multiplier that determines the current market value of the validator's pbluechips. The current market value of pbluechips can be found at an on-chain oracle existing on the BlueChip blockchain.

Trust Scores are calculated using a weighted snapshot of the validator's performance over time. The most recent behavior is emphasized more heavily, with historical behavior decaying in influence. The scoring window is divided into six trust windows, each lasting five epochs (~15 days). After each window, its influence decays according to a fixed weighting schedule.

Once a validator's effective Trust Score is calculated, the following equation is used to determine its pbluechip multiplier:

$$Multiplier = 1 + (0.25 * trust_score/10)$$

This caps the multiplier at 1.25 (for a perfect score of 10.0) and bottoms out at 1.00 (neutral trust baseline). A multiplier above 1.00 indicates strong validator performance and increases the trade value of that validator's pbluechips in bubble transactions.

The Trust Score is composed of the following metrics, each contributing to a validator's score at each update. These inputs are evaluated per scoring window:

Positive Trust Metric	**Trigger Condition**	**Score Impact**
RLF / Liquidity Vault ratio	Ratio > 1.0	+1.0
Raw RLF deposit size	Deposit above threshold	+0.5
Bubble LTV	Stayed below 70%	+1.0
LP volatility	LP value stayed stable	+0.5
% fulfilled unbond requests	≥99% fulfilled	+0.5
Delegator count trend	Upward trend	+0.5
Epochs since last negative event	3 or more epochs with no violations	+0.5
Epochs actively contributing liquidity	3 or more active epochs	+0.5
Consecutive good epochs	3+ penalty-free epochs	+0.5
Consecutive good epochs	5+penalty-free epochs	+0.5
Weight distribution across pools	Well-balanced (no pool > 60%)	+0.5
Bubble rebalancing frequency	Rebalanced at least once in last 3 epochs	+0.5
% self-stake	≥10% of total stake is self-bonded	+0.5
Validator commission changes	Commission stayed stable	+0.5

Negative Trust Metric	Trigger Condition	Score Impact
Last missed loan	Missed a loan payment	−1.5
Overconcentration in 1 pool	≥70% of liquidity in a single pool	-0.5
Bubble turnover frequency	Swapped >30% of bubble assets in a single epoch	-0.5
Validator not providing liquidity	0 liquidity pool positions in >5 consecutive epochs	-0.5
Validator parameter churn	≥3 unbonding or commission changes in 10 epochs	-1.0
Rapid validator join/exit pattern	In/Out of active validator set >3 times in 6 epochs	-0.5
Last missed penalty	Received a penalty	−1.0
Last withdrawal failure	Delegator withdrawal failed	−1.0
Last liquidation	Was liquidated	−3.0
Unbond delay occurrences	Multiple delays during epoch	−0.5
Near-miss liquidation events	Bubble LTV ≥ 89% at any time	−1.0
Bubble valuation shocks	Bubble dropped >20% in a single epoch	−0.5

The final Trust Score is calculated as a weighted average of the last six scoring windows. More recent windows contribute more heavily:

Value Window (t-i)	Window Score	Decay Weight	Weighted Contribution
t-0	6.4	0.40	2.560
t-1	6.8	0.25	1.700
t-2	7.2	0.15	1.080
t-3	7.0	0.10	0.700

t-4	6.5	0.05	0.325
t-5	5.8	0.05	0.290
→ **Effective Score**	-	-	**6.655**
→ **Final Multiplier**	-	-	**1.16**

A final important note regarding pbluechip pricing is how this price will be derived in the future. CLOBs are a new focus in the crypto space, and they are perfect for this situation. Setting up a CLOB, or connecting to another will take loads of work. Starting with a simple multiplier is where the pricing will begin.

Composite Score vs Security Contributions

Although it is included in a validator's composite score, the liquidity provided to pools through the Liquidity Vault is not necessarily considered when choosing which validators maintain the chain for rewards. If it were, by simply choosing not to use the Liquidity Vault, a validator could easily obtain a majority stake of chain security relative to the rest of the validator set. For instance, let's say there are 5 validators, and the required staking amount is 5 tokens. 4 validators could all have 5 tokens staked to the chain with 30 each to the Liquidity Vault. Their score would be 35 each and a total 25 staked to the chain between them. The fifth validator could have 26 staked to the chain and 0 in his Liquidity Vault. So even though the fifth validator has the lowest score, they control approximately 51% of the network.

A second reason the choice was made to not always count the Liquidity Vault value is the inevitableness of it causing sporadic security for the chain. If the Liquidity Vault were to be included, the price swings of assets in liquidity pools would drastically affect the safety of the protocol, not simply returns. In a vacuum, this decision disincentivizes validators from utilizing their Liquidity Vault. To mitigate this, there is a chain staking limit from validators simply named the **stake cap.** The stake cap is a hard limit on how much validators can stake directly to the chain. If a validator would like to improve their overall composite score after the stake cap is reached, they have to provide liquidity to their vault.

Validator Set Checks

There are two important time periods when considering the stake rewards for the validators. The first is a simple check to see which validators are staking the appropriate amount. The reason this is a simple check is due to the validator stake cap.

First, a check is done to see if each validator is still providing the required amount of stake. If each current validator is, a check is done to see if their stake is the stake cap. If each validator's stake is the stake cap, the check is complete. If the amount is not the stake cap, a validator

outside the current set may be grabbed. The validator is chosen by the amount of bluechips it would be able to stake immediately. This amount must be higher than the lowest current validating node.

If there are multiple validators with the same amount staked and are above the lowest current staking validator (ie, multiple non validating nodes are staking the validator stake cap) one is chosen at random. It does not take into consideration the validator's composite score, which includes its liquidity provisions. As many as all the validators can be replaced at once if enough validating nodes outside the current set are prepared to stake more. A simple check takes place every 24 hours (14,400 blocks).

The second check is a complex check. This check *potentially* takes the composite score into consideration. This check is complex because it takes into consideration all validators with the potential to stake a valid amount.

A complex check does a simple check first, looking for the highest amount staked to the chain by a validator. From there, it looks for validators in descending order up to the required amount of validators needed to validate the chain and stops. The one caveat is, if there are validators staking the same amount at the end of the list. This can happen if more nodes are able to stake the validator stake cap than there are validating slots, or if at the end of the list, multiple nodes happen to be staking the exact same amount. If this circumstance arises, the validating node with the highest composite score is selected. As many as all the validators can be replaced at once if enough validating nodes outside the current set have a higher composite score. This check takes place every 5 days (72,000 blocks).

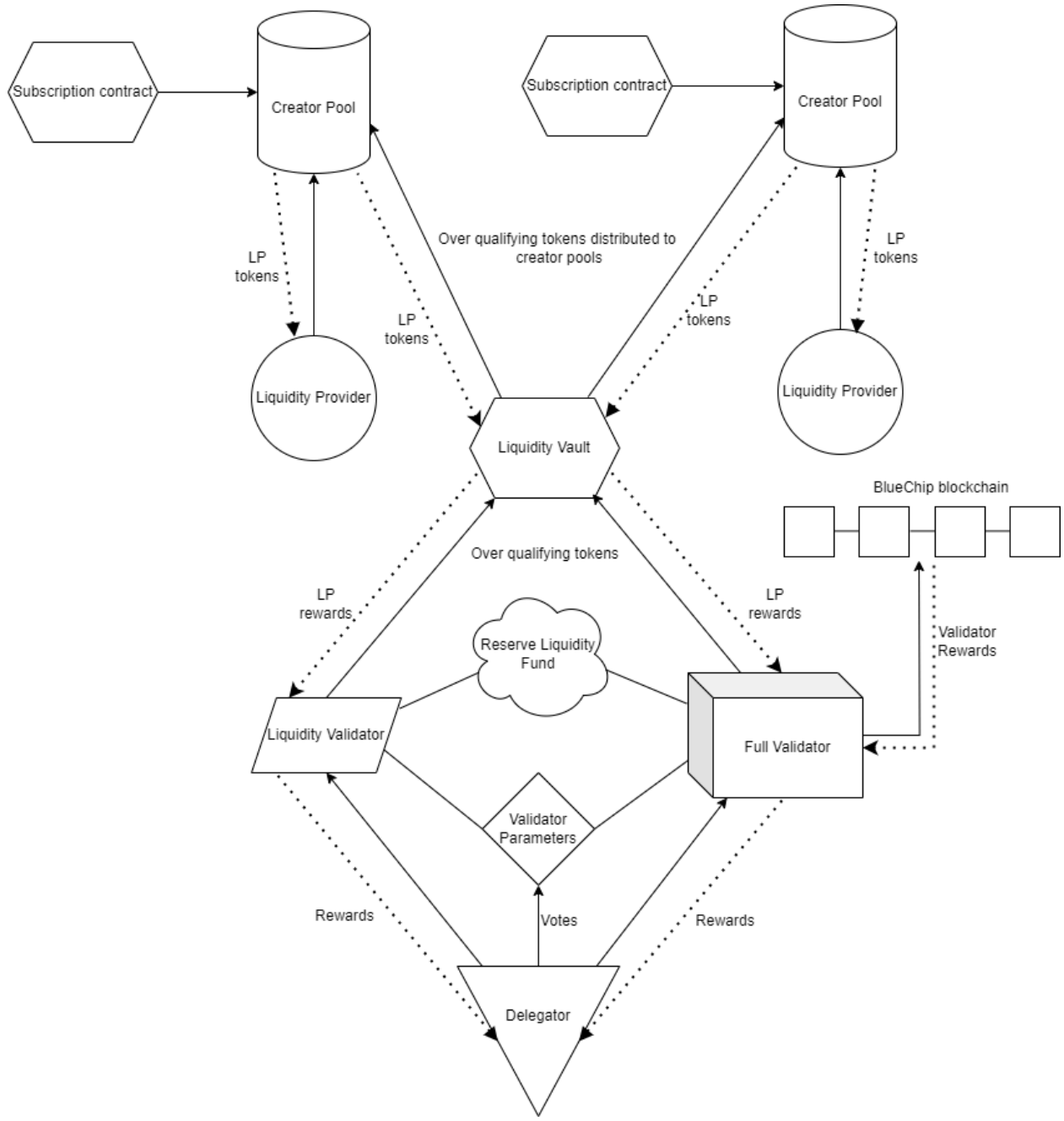
<u>Composite Score</u>

How is the composite score… scored? The composite score is the amount staked to the chain directly, plus the value in the Liquidity Vault. Between each complex check there are 6 value posts by the validator's Liquidity Vault, relaying the Liquidity Vault's total value. Value posts are pseudo random to help mitigate the gaming of the posts. Once the 6 value posts have been made, the median score will be used.

<u>Conclusion</u>

The BlueChip protocol assigns each creator or entity a dedicated crypto token and liquidity pool. However, without sufficient liquidity, these tokens risk becoming unstable, and pool efficiency suffers, potentially undermining the stability of the BlueChip token itself. This protocol upgrade introduces mechanisms to inject liquidity more sustainably while reducing common downsides for liquidity providers. By enabling validators to direct capital into liquidity pools through their dedicated Liquidity Vaults, BlueChip enhances both liquidity and validator incentives. These vaults not only distribute LP rewards to validators and their delegators but also contribute to the

validator's composite score. To further support liquidity under stress, the Reserve Liquidity Fund (RLF) acts as a backstop — offering loans, penalties, and incentives that reinforce responsible validator behavior. Going forward, additional refinements will aim to improve validator experience, safeguard liquidity, and build a decentralized economy where creators and investors alike benefit from healthy, resilient liquidity pools.



<u>How Do We Get There?</u>

Introduce liquidity vaults

- ☐ The first steps of the RLF and liquidity providing protocol is the liquidity providing itself. Being the soul of the whole chain upgrade, getting the liquidity providing right is crucial. In the initial upgrade, validators can provide liquidity to their liquidity vault and obtain rewards from pools in the ecosystem. These rewards can be passed along to their delegators. The initial upgrade introduces the “composite score” and the “stake cap”.
- ☐ Important note - the only parameter the validator can alter at this point is the percent of rewards passed along to delegators. They can not set withdrawal periods or any similar parameters at this time. The rewards would be passed along based on the % set by the validator and then distributed to delegators based on their stake.

Introduce the RLF

- ☐ The second upgrade to the system is significant. The addition of the reserve liquidity fund brings with it many capabilities for the validator. The validator can now deposit into, and take loans from, the reserve liquidity fund based on their deposit. What this also means is, the validator can now set parameters such as withdrawal periods for delegator liquidity.

Introduce the bubble and pbluechips

- ☐ Once the RLF is established and working properly, we can take next steps and begin making the system truly unique. Introducing the bubble and validator specific pbluechips is the focus of this stage of the upgrade. The bubble represents 15% of deposits and the validators are able to borrow 85% of their full RLF deposit (including bubble). This upgrade also introduces bubble trading. All trades to and from the bubble must be 1:1 and the validator can only trade their pbluechips for bluechips. They can not accumulate creator tokens or other tokens natively created on the BlueChip blockchain. The price oracle for validator pbluechips is also brought into the fold and validators actions will now have an impact on their pbluechips price. Existing RLF deposits will be migrated via an in-place wrapper-mint during the upgrade handler.

Introduce bubble trading with native tokens ONLY.

- ☐ Once the functionality of the bubble has been determined safe, trading opportunities begin to open up for validators. They can now make trades from their bubble for any native asset to the bluechip blockchain. This includes other creator tokens, other pbluechips, and ofcourse the bluechip itself. 1:1 trading still applies.
- ☐ It is important to note this is not one overhaul upgrade. It most likely occurs in increments. For instance, 5% of bubbles can be unique assets, followed by 10% etc. The ultimate completion of this stage is when the bubble can be fully unique assets.

Introduce bubble trading with any token including off chain assets.

- ☐ Further down the line, the bubble will begin accepting all tokens. This of course will be a slow bleed and will most likely not be one large installment. Different assets from different chains will require different compatibility requirements. Moreover, some of the more prominent tokens with direct interoperability with Cosmos, like Bitcoin and ETH, will be prioritized.